

Technical Data Sheet	Complex multifunctional admixture D5 for concrete HS Code 3824 40 000 0	Total pages 3	16.01.2026
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1. PRODUCT IDENTIFICATION

Technical name	Admixture D5
Description	The admixture D5 is a finely-milled powder of beige color and was designed to obtain waterproof high-strength concretes and mortars with accelerated strength gain and extremely high sulphate-resistance. The admixture D5 is produced on the basis of natural mineral raw materials, which have active pozzolanas, specially selected finely ground rocks and a superplasticizer as components.
Packing	15 kg paper valve bags, 1000 kg big-bags or other packaging on the customer's request
Available form	Powder form. Bulk density is 750-800 kg/m ³
Shelf life	36 months
Regulatory document for production	TS (Technical Specifications) 5745-002-37415339-2015
Standard compliance	Admixture D5 complies with BS EN 934-2, BS EN 480-1 (T.3.1), BS EN 12390-8.

2. MANUFACTURER AND/OR SUPPLIER INFORMATION

Name of the manufacturer	Scientific and Production Enterprise TOKAR Co. Ltd
Address	19 A Pozharskogo str., Vladikavkaz, 362002, Republic of North Ossetia-Alania, Russia
Phones	+7(938)883-10-37, +7(918)823-08-05
E-mail	info.dobavkad5@mail.ru

3. KEY BENEFITS SUMMARY

High water-resistance	Concretes with the admixture D5 acquire the highest possible water resistance (up to 20 Bar and more), which allows to exclude completely any additional waterproofing of concretes in water-saturated soils or directly in water. High water resistance of concretes with the admixture D5 is maintained for the entire life of a structure.
High sulfate-resistance	Concretes with the admixture D5 acquire high sulfate-resistance even with ordinary Portland cement. This allows to avoid any additional waterproofing of concrete even when concrete is operated in aggressive environment, including sea water, oil and oil products.
High strength	The use of the admixture D5 increases the concrete strength at the age of 28 days by an average of 30% compared to concretes without admixtures (in concrete mixes with equal slump), or reduces cement content by 15-20%.
Acceleration of strength gain	The use of the admixture D5 in concretes accelerates the strength gain of concretes on the first days of hardening not less than by 2 times.
Plasticizing characteristics	The admixture D5 increases flowability of concrete mixes from P1 to P5 or reduces water consumption by 15-25% in mixes with

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	equal slump. When using the admixture D5 there is no need in any other plasticizers.
Self-sealing of through cracks	D5 has properties of self-sealing of through cracks up to 0.5 mm wide, which can appear in concrete structures during exploitation due to different dynamic loads.
Eco-friendliness	The admixture D5 does not contain components which can affect negatively cement stone and steel reinforcement, it doesn't form toxic compounds in atmosphere and water. The admixture D5 is approved for producing concrete and reinforced concrete structures that are in contact with drinking water.
High frost-resistance	The use of D5 increases frost-resistance of concretes not less than by 2 grades and allows to get concretes with frost-resistance of F400 and more.
Additional properties	The admixture D5 is not aggressive towards reinforced steel, prevents it from corrosion, enhances passivating characteristics of concrete towards reinforced steel and is recommended for use in reinforced concrete elements of essential structures. The admixture D5 increases adhesion between concrete and reinforcement by 36%. The admixture D5 increases adhesion to old concrete by 1.9 times or by 90% as compared to adhesion of concrete without admixtures.

4. PROPERTIES

Water reduction EN 480-1 (t. 3.1)	≥ 12% compared with Reference mix
Increase of consistence	Increase ≥ 160 mm from initial flow
Retention of consistence EN 480-1	At 30 mins ≥ Reference mix at initial
Compressive strength EN 480-1 (t. 3.1)	Fulfilled
Air content EN 480-1 (t. 3.1)	Fulfilled. Less than 2% additional air is entrained at normal dosages
Corrosion behaviour	Contains components only from EN 934-1: Annex A.1

5. COMPONENTS INFORMATION

Sodium oxide (Na ₂ O)	1,04%
Potassium oxide (K ₂ O)	1,09%
Chlorine ion (Cl ⁻ , comm.)	0,065%

6. RECOMMENDATIONS FOR USE

Dosage	The recommended dosage of the admixture D5 is 2-3% by the total weight of cement.
Preliminary tests	Before using the admixture at plants of precast reinforced concrete or at concrete mixing stations, it is necessary to conduct preliminary tests under laboratory conditions in order to determine its effectiveness when interacting with cements and aggregates that are used at each particular enterprise.
Preparation	When concrete mix is being produced, the admixture D5 is

Mixing time

introduced into a concrete mixer in dry form together with sand or cement. There is no need to pre-mix the admixture D5 with water!

Mixing time is 30-90 seconds (depending on the type of mixer and the required flowability of concrete mix) and, within this time, the admixture D5 is evenly distributed throughout the concrete mix. As the admixture D5 contains a superplasticizer, therefore this fact must be kept in mind when determining the amount of water in concrete mix.

Treating «cold» joints

In concretes with the admixture D5 «cold» joints do not leak under the water pressure of up to 12 Bar (1,2 MPa). To provide that, it is necessary to perform the following steps:

- the surface of previously laid concrete (with the admixture D5) should be cleaned from the cement crust immediately after hardening (6-8 hours after pouring). It is necessary to expose the concrete structure, wash off dust and dirt and abundantly water the concrete in the place of the future «cold» joint;
- after breaks in concreting (which can last from several days to several weeks), just before laying freshly-mixed concrete with D5, it is necessary to wash the place of the «cold» joint again to wash off dust and dirt from the surface of earlier stripped concrete;
- under these circumstances, high adhesion and fusion at the molecular level between the hardened and freshly laid concrete with the admixture D5 will be provided and high waterproofing of «cold» technological joints (both vertical and horizontal) will be achieved.

«Cold» joints in concretes with D5 don't need any additional waterproofing provided that water pressure on «cold» joints will not exceed 12 Bars (1,2 MPa).

7. APPLICATION

- Drinking water tanks
- Swimming pools
- Oil-storage tanks
- Treatment facilities
- Underground parking
- Underground and underwater tunnels
- Fish-breeding pools
- Floating platforms etc.

8. CERTIFICATES

- Certificate of conformity of the Russian Federation
- Certificate of conformity of New Zealand
- ISO certificate of conformity
- Certificate of State Registration
- Certificate of radiation quality
- Ecological Certificate

